

TSUNAMI ADVISORS · AI INTERN CASE STUDY

Build Documentation

TariffMapper — automated China ↔ Indonesia customs tariff code mapping (prototype).

GPT-4o

Next.js 16

TypeScript

HS 2022

CCC 2024

BTKI / AHTN

Optional Supabase

Status: MVP complete · **Version:** 1.0 · **Date:** May 2026

Deliverable: Build documentation (2–4 pages equivalent) per case study specification.

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1 Project overview

TariffMapper is a generative-AI-first prototype that maps import and export product classification codes between China and Indonesia. The user enters a product description, HS code, or national tariff code; the system returns the top five likely matches in the *target* country with confidence, match type, tariff indicator, explanation, and source references.

The system is designed to support:

- **One-to-many mappings** — one HS anchor can expand to several national codes.
- **Ambiguous descriptions** — natural language is resolved to plausible HS anchors with calibrated confidence.
- **National divergence** — differences beyond six digits (China CCC vs Indonesia BTKI/AHTN) are surfaced in `divergenceNote`.
- **Low certainty** — results below the rubric threshold use `manual_review` and are called out in the UI.

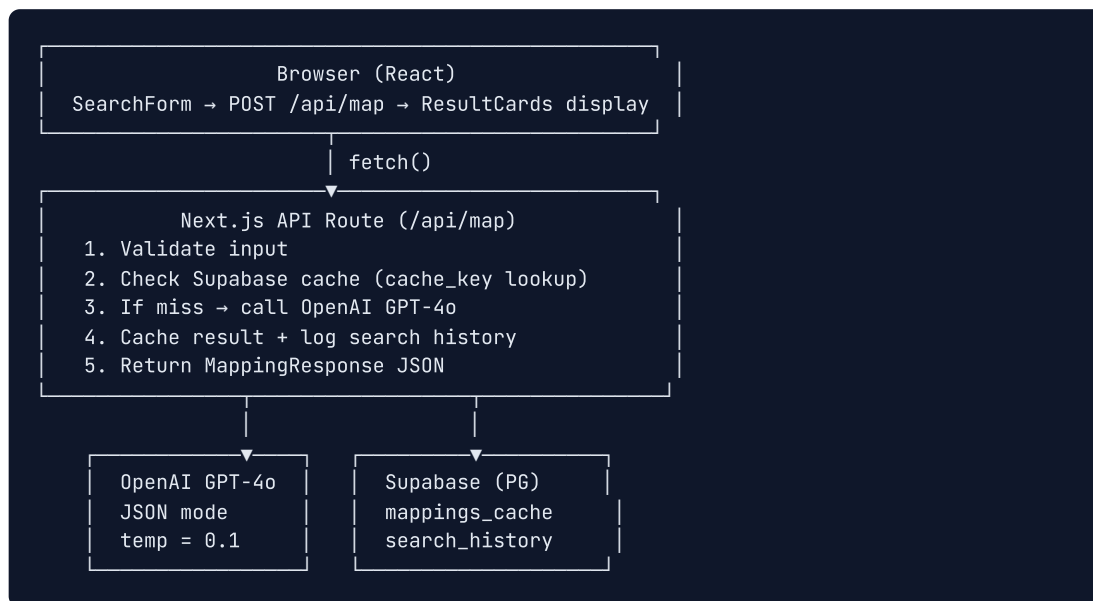
2 Tools and technology stack

Layer	Tool / library	Purpose
AI engine	OpenAI GPT-4o	Mapping, semantic interpretation, explanations
Frontend	Next.js 16 (App Router)	React UI, static + server routes
Styling	Tailwind CSS	Responsive layout
Language	TypeScript	End-to-end type safety
API	Next.js route handlers	<code>POST /api/map</code>
Persistence (optional)	Supabase (PostgreSQL)	Result cache + search history
Hosting	Vercel (recommended)	Zero-config Next.js deploy

Why GPT-4o?

1. No public, complete bilateral CCC↔BTKI crosswalk exists — a pure database approach would require building and maintaining a proprietary crosswalk.
2. Product descriptions need semantic understanding, not only exact string match.
3. `response_format: json_object` plus low temperature (0.1) yields consistent structured JSON.
4. The model can articulate *why* a code was suggested (traceability requirement).

3 System architecture



Request lifecycle: (1) User submits query, direction, and search mode. (2) API checks cache key `direction::normalised_query`. (3) On miss, system + user prompts are sent to GPT-4o. (4) Response is parsed; cache and history writes are non-blocking. (5) Client renders cards with confidence, basis, and expandable details.

4 Key prompts and prompt engineering

4.1 System prompt design

The system prompt encodes HS 2022 structure, China CCC 2024 extensions, Indonesia BTKI 2022 (AHTN-aligned), and a four-tier confidence rubric. It enforces: no fabricated rates; national divergence notes; `manual_review` below 40%; and four `matchBasis` values: `hs_digits`, `semantic`, `tariff_structure`, `ahtn_extension`.

Tier	Score	Label	Criteria
1	85–100%	exact	Strong HS anchor + clear target national extension
2	65–84%	likely	HS match; minor sub-heading ambiguity
3	40–64%	partial	Adjacent HS / chapter-level only
4	0–39%	manual_review	Multi-chapter ambiguity or insufficient data

4.2 Prompt iterations (v1 → v4)

- **v1** — Free-text output; fabricated rates.
- **v2** — JSON mode + anti-fabrication rules; scores miscalibrated (too high).
- **v3** — Tiered rubric + citations; anchor vs national codes still conflated.
- **v4** — Anchor-first instruction; `matchBasis` + `divergenceNote`; temperature 0.1.

4.3 Direction enforcement (refinement). The system prompt is now built per request via `buildSystemPrompt(direction)`. A **CRITICAL DIRECTION RULE** block states that every `matches[].code` must be a *target-country* national code (Indonesia BTKI when mapping China→Indonesia; China CCC when mapping Indonesia→China). This fixed cases where the model echoed source-country codes or treated direction ambiguously. Identical 10-digit strings can still appear for both countries on some commodities (e.g. crude palm oil) — that is noted in `processingNote` when applicable.

5 Data sources and selection rationale

Source	Reference	Why selected
WCO HS 2022	wcoomd.org (HS Nomenclature 2022)	International six-digit anchor for all mappings
China CCC 2024	tariff.customs.gov.cn	Official eight–ten digit extensions and MFN context
Indonesia BTKI 2022	insw.go.id / beacukai.go.id	Official tariff book; AHTN-based national codes
ASEAN AHTN 2022	asean.org	Framework for Indonesian eight-digit structure
ACFTA / RCEP	Official FTA schedules	Preferential rate context where relevant

Only **public** sources and model knowledge within assignment constraints — no non-public customs databases or invented legal certainty.

6 Matching logic and ranking methodology

```
Input (description / HS code / local code)
|
▼ Step 1 – Anchor resolution (6-digit HS)
▼ Step 2 – Map to target national codes (8-10 digits)
▼ Step 3 – One-to-many expansion where the schedule branches
▼ Step 4 – Score with rubric; sort descending by confidence
▼ Step 5 – Divergence check → populate divergenceNote
```

6.1 Fuzzy or incomplete product descriptions

For vague inputs (e.g. “industrial pump”), the pipeline returns candidates across plausible chapters, lowers confidence, and explains ambiguity in `processingNote` so users see *why* the spread exists.

6.2 National codes beyond six digits

Examples documented in the build: Chapter 85 (China often more granular on electrical sub-lines); Chapter 15 (Indonesia palm-oil granularity); Chapter 84 (automation equipment

divergence at eight digits). Each result can carry a `divergenceNote` tying divergence to the two national schedules.

7 Supabase schema (optional infrastructure)

```
-- mappings_cache: dedupe identical queries per direction
CREATE TABLE mappings_cache (
  id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
  cache_key TEXT UNIQUE NOT NULL,
  result JSONB NOT NULL,
  updated_at TIMESTAMPTZ DEFAULT NOW()
);

-- search_history: lightweight audit / analytics
CREATE TABLE search_history (
  id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
  query TEXT NOT NULL,
  direction TEXT NOT NULL,
  search_mode TEXT NOT NULL,
  hs_anchor TEXT,
  result_count INTEGER,
  created_at TIMESTAMPTZ DEFAULT NOW()
);
```

Note: The live app runs without Supabase; caching and history are skipped if env vars are absent.

8 Limitations and unresolved classification risks

Hard limitations: model knowledge cutoff; no live MFN/FTA rate API in MVP; no binding ruling authority; ten-digit micro-codes (e.g. CIQ-driven splits) may be incomplete without a live tariff database.

Unresolved risks: dual-use goods and licensing; Chapter 98 / special provisions; schedule drift between HS amendments and national sync; highly ambiguous multi-function products.

9 What AI did well vs what requires human judgment

AI strengths	Assessment
Semantic anchoring	Strong on typical product language → HS headings
One-to-many suggestions	Reliably produces five ranked candidates
Explainability	Natural-language rationale + basis labels
Ambiguity signalling	<code>manual_review</code> tier used for low-confidence cases

Human judgment still required	Reason
Exact duty / FTA eligibility	Requires live official schedule + origin documentation
Ten-digit precision	Product-specific attributes and national sub-notes
Regulatory overlays	Licences, SPS, certification — outside pure HS mapping
Binding classification	Only customs authorities issue advance rulings

10 Deployment (summary)

Required: `OPENAI_API_KEY` in `.env.local` (or Vercel env). **Optional:**

`NEXT_PUBLIC_SUPABASE_URL` and `NEXT_PUBLIC_SUPABASE_ANON_KEY` for cache + history.
Typical cost: roughly \$0.01–0.03 per mapping call depending on token usage.

Local: `cd tariff-mapper && npm install && npm run dev`. Production: `vercel deploy` with the same environment variables.

11 Reflection

What worked best: JSON-mode GPT-4o with a domain-heavy system prompt produced stable structured payloads suitable for direct UI binding; optional Supabase cache reduces repeat OpenAI cost for demos.

Where matching was weakest: multi-chapter consumer electronics (“smart home” style queries); highly granular Indonesian palm fractions vs Chinese line splits; dense chemical/pharma trees with many national children.

Next version: live rate ingestion from official portals; curated or embedded crosswalk / embeddings; HS tree browser; CSV batch mode; analytics on low-confidence clusters from `search_history`.